

4.13.46

EE25BTECH11042 - Nipun Dasari

Question:

A straight line L is perpendicular to the line $5x - y + 1 = 0$. The area of the triangle formed by L and the coordinate axes is 5. Find the equation of the line L .

Solution:

Let the given line be L_1 and the required perpendicular line be L . The equation of the given line L_1 is $5x - y = -1$. Using the matrix form $\mathbf{n}^\top \mathbf{x} = c$, its parameters are:

$$\mathbf{n}_1 = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad c_1 = -1 \quad (0.1)$$

Let the required line L have the equation $\mathbf{n}_2^\top \mathbf{x} = c_2$. Since the lines are perpendicular, the dot product of their normal vectors must be zero, so $\mathbf{n}_1^\top \mathbf{n}_2 = 0$.

Let's assume the normal vector for line L is $\mathbf{n}_2 = \begin{pmatrix} 1 \\ m \end{pmatrix}$. We can solve for m :

$$\mathbf{n}_1^\top \mathbf{n}_2 = \begin{pmatrix} 5 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ m \end{pmatrix} = 5 - m = 0 \quad (0.2)$$

$$\implies m = 5 \quad (0.3)$$

Therefore, the normal vector for line L is $\mathbf{n}_2 = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$, and its equation is $x + 5y = c_2$.

The x-intercept, $\mathbf{a}$, is the intersection of line L and the x-axis ($y = 0$). We solve the system by reducing the augmented matrix to RREF:

$$\left(\begin{array}{cc|c} 1 & 5 & c_2 \\ 0 & 1 & 0 \end{array} \right) \xleftrightarrow{R_1 \leftrightarrow R_1 - 5R_2} \left(\begin{array}{cc|c} 1 & 0 & c_2 \\ 0 & 1 & 0 \end{array} \right) \quad (0.4)$$

This gives the solution $x = c_2$ and $y = 0$. So, the x-intercept vector is $\mathbf{a} = \begin{pmatrix} c_2 \\ 0 \end{pmatrix}$.

The y-intercept, $\mathbf{b}$, is the intersection of line L and the y-axis ($x = 0$). We solve by reducing the augmented matrix to RREF:

$$\left(\begin{array}{cc|c} 1 & 5 & c_2 \\ 1 & 0 & 0 \end{array} \right) \xleftrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 1 & 5 & c_2 \end{array} \right) \quad (0.5)$$

$$\xleftrightarrow{R_2 \rightarrow R_2 - R_1} \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 5 & c_2 \end{array} \right) \quad (0.6)$$

$$\xleftrightarrow{R_2 \rightarrow \frac{1}{5}R_2} \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & c_2/5 \end{array} \right) \quad (0.7)$$

This gives the solution $x = 0$ and $y = c_2/5$. Thus, the y-intercept vector is $\mathbf{b} = \begin{pmatrix} 0 \\ c_2/5 \end{pmatrix}$.

The area of the triangle formed by the origin, $\mathbf{a}$, and $\mathbf{b}$ is 5. Since $\mathbf{a}$ and $\mathbf{b}$ lie on the axes, the angle θ between them is 90° .

$$\text{Area} = \frac{1}{2} \|\mathbf{a} \times \mathbf{b}\| \quad (0.8)$$

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin(\theta) \quad (0.9)$$

$$\Rightarrow \frac{1}{2} \|\mathbf{a}\| \|\mathbf{b}\| \sin(90^\circ) = 5 \quad (0.10)$$

$$\Rightarrow \frac{1}{2} \left(\sqrt{c_2^2 + 0^2} \right) \left(\sqrt{0^2 + (c_2/5)^2} \right) (1) = 5 \quad (0.11)$$

$$\Rightarrow \frac{1}{2} |c_2| \left| \frac{c_2}{5} \right| = 5 \quad (0.12)$$

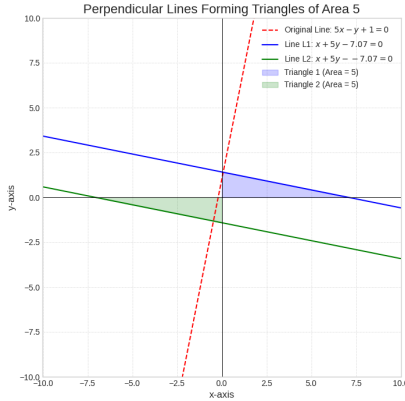
$$\Rightarrow \frac{|c_2^2|}{10} = 5 \quad (0.13)$$

$$c_2^2 = 50 \Rightarrow c_2 = \pm \sqrt{50} = \pm 5\sqrt{2} \quad (0.14)$$

There are two possible lines that satisfy the given conditions. Substituting the values of c_2 into the matrix equation $\mathbf{n}_2^\top \mathbf{x} = c_2$, the possible equations for line L are:

$$\begin{pmatrix} 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 5\sqrt{2} \quad (0.15)$$

$$\begin{pmatrix} 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -5\sqrt{2} \quad (0.16)$$



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